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DePeak

Introduction

DePeak is a tool for applying diversity effects to load profiles. When individual building load profiles are simply summed, the resulting peak overstates the actual simultaneous demand, since not all buildings reach their peak load at the same time. DePeak addresses this by reducing the peak of an aggregated profile according to a diversity factor (DF), while preserving the total energy demand.

Two independent approaches are implemented:

- The Winter approach scales the profile toward a DF that is either set directly or calculated from the empirical approximation formula by Winter et al. (2001) [1], based on the number of buildings.
- The Shift approach models the diversity effect through a different, physically motivated mechanism: individual building profiles are shifted in time before summation, which reduces the peak of the summed curve through decorrelation.

The two approaches are independent of each other and can be used separately, depending on the input data available and the mechanism the user wants to investigate. Chapters 1 and 2 describe the methodology, results, and limitations of each approach in detail.

1 Winter Approach

This section describes the diversity-factor approach developed as part of the simultaneity analysis, which applies a DF to an aggregated, non-coincident load profile that results from summing multiple building profiles without accounting for a simultaneity effect. The DF itself is either given directly or calculated using the empirical approximation formula by Winter et al. (2001) [1].

The goal is to produce an adjusted load profile that satisfies two boundary conditions derived from the definition of the simultaneity factor: the peak is, by definition, reduced by exactly the DF, while the total energy demand must remain unchanged - accounting for simultaneity only changes the temporal distribution of the load, not the total amount of energy required. These two conditions align with the requirements publicly documented by nPro [2], which served here as additional orientation:

- Boundary condition 1: $new\ peak = original\ peak \cdot DF$
- Boundary condition 2: the annual energy (area under the curve) is preserved exactly

These two boundary conditions served as the conceptual starting point; the internal algorithm nPro uses to implement the energy redistribution is not publicly documented and was not the subject of this work. The approach developed here is an independent implementation: the tool first redistributes the energy removed by peak clipping within the section between the two neighboring minima that contains the respective peak; if this is not sufficient to exactly satisfy both boundary conditions, a supplementary correction is applied across the entire profile.

1.1 Theoretical Basis

The DF for n simultaneously considered buildings is calculated using the empirical approximation formula by Winter et al. (2001) [1]:

$$DF = a + \frac{b}{1 + \left(\frac{n}{c}\right)^d}$$

Here, a , b , c , and d are empirically fitted parameters. In Winter's original work:

Table 1

Parameter	Winter's Original Value	Meaning
a	0.449677646267461	Lower asymptotic bound of the DF for a very large number of buildings
b	0.551234688	Distance between the lower bound a and the DF for a very small number of buildings
c	53.84382392	Characteristic number of buildings / half-saturation parameter of the function
d	1.762743268	Shape parameter that determines the curvature and steepness of the transition

Winter's formula was calibrated on combined heating + domestic hot water load measured at the customer substation level (typical for district heating systems, which meter total heat draw per building rather than heating and DHW separately)

In the tool, only the parameter a remains freely selectable; c indicates the order of magnitude of the number of buildings at which the DF is roughly halfway between its starting value and its limit value and is fixed at Winter's original value. d is likewise fixed at Winter's original value. For b , Winter's fixed original value is not adopted; instead, $b = 1 - a$ is deliberately set.

Winter's a and b were jointly fitted such that $a + b = 1$ holds, which ensures the boundary condition $DF(1) = 1$ (with only one building, no simultaneity reduction should occur). If only a were freely changed while b remained at Winter's fixed original value, $DF(n) > 1$ could result - an unphysical increase rather than a reduction of the load.

Winter's formula was validated empirically only for $1 < n \leq 200$. For $n > 200$, the calculation is continued beyond the originally defined range; the tool automatically issues a note in this case. Winter et al. themselves tested this extrapolation on a real network with $n = 255$ (formula: $DF \approx 0.47$, measured: $DF = 0.51$) and rated it "plausible" - though this is not a guarantee [1].

1.2 Methodology

1.2.1 Local Peak Limiting and Energy Redistribution

First all local maxima (peaks) and minima (valleys) of the aggregated profile are identified. A point counts as a peak or valley if its deviation from the nearest valley or peak is at least 2% of the total value range (maximum minus minimum) of the profile - smaller fluctuations are ignored as noise.

For each local peak, the value is multiplied by the DF; the “clipped” energy is redistributed proportionally to the available headroom (distance to the new, capped peak value) across the remaining hours of the same local section - bounded by the two neighboring valleys on the left and right. This keeps the energy redistribution spatially confined to the immediate temporal context of the respective peak, rather than diluting it across the entire time series.

1.2.2 Global Peak Correction

Since the local redistribution in Step 1 can, in individual cases, cause time steps outside the original peaks to exceed the global target peak (peak of the entire time series $\times$ DF), such time steps are capped at the target peak; the excess energy is redistributed proportionally across all time steps below the target peak.

1.2.3 Exact Conservation of Total Energy

Finally, any remaining difference between the total energy of the original profile and the adjusted profile is distributed proportionally across all time steps below the target peak. If this again leads to exceedances of the target peak, these are capped again and the excess energy redistributed, until boundary condition 2 is satisfied within a tolerance of 10^{-6} .

This approach, too, is a plausible approximation, not a confirmed reconstruction of nPro's internal algorithm, which is not publicly documented.

1.3 Results

The following examines whether the implemented approach correctly applies the given DF to different starting profiles (23-building profile with $\alpha = 0.6$ and 10-building profile with $\alpha = 0.6$) while satisfying the two essential boundary conditions: the reduction of the annual peak according to the given DF, and the exact conservation of annual energy. In addition, it was examined how the approach behaves with different input data formats.

Building load profiles generated with nPro [3] were used as input data. Two cases were examined for this purpose: a 23-building profile with an already aggregated sum profile, and a 10-building case with ten separate individual profiles. In the first case, the DF was applied directly to the existing sum profile. In the second case, the individual profiles were first aggregated by the tool into a sum profile and then adjusted with the given DF.

The main results of both cases are summarized in Table 2:

Table 2

Metric	23 buildings	10 buildings
DF	0.9270	0.9804

Peak, raw (kW)	1,757.7	78.7
Peak, with DF (kW)	1,629.4	77.2
Peak reduction (kW / %)	128.3 / 7.30 %	1.5 / 1.96 %
Annual total, raw (kWh)	2,433,035.1	67,764.0
Annual total, with DF (kWh)	2,433,035.1	67,764.0

The temporal effects of the adjustment are shown in Figures 1-4. The annual time series shows the original and the DF-adjusted profile respectively; the detailed views show the area around the respective annual peak.

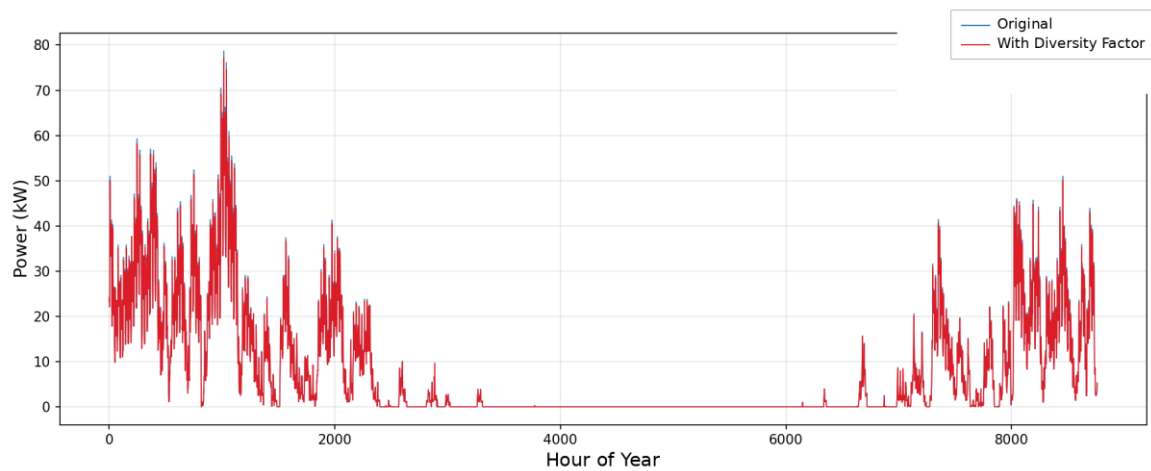


Figure 1: 10-building district - annual time series, original vs. with DF ($a = 0.6$).

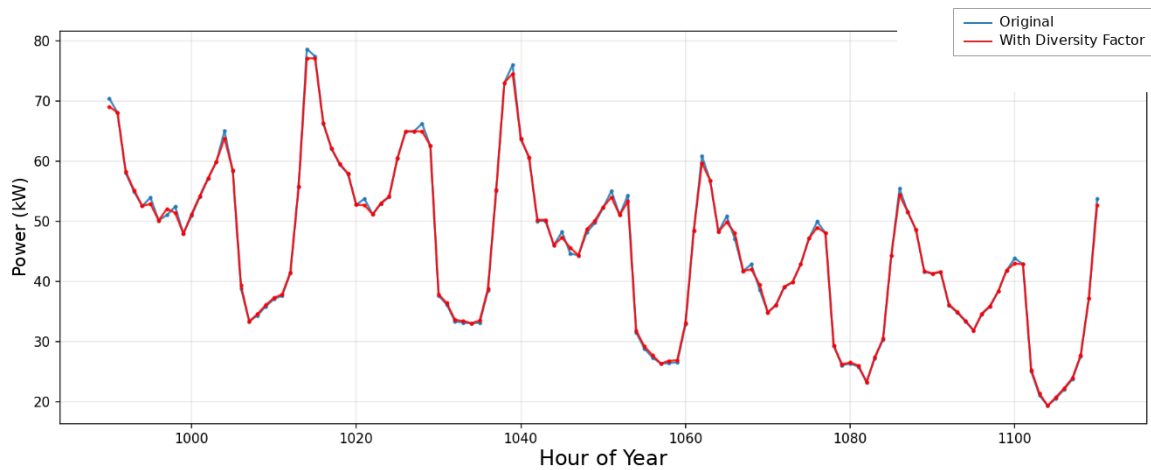


Figure 2: 10-building district - detail view around the annual peak (hour 990-1110).

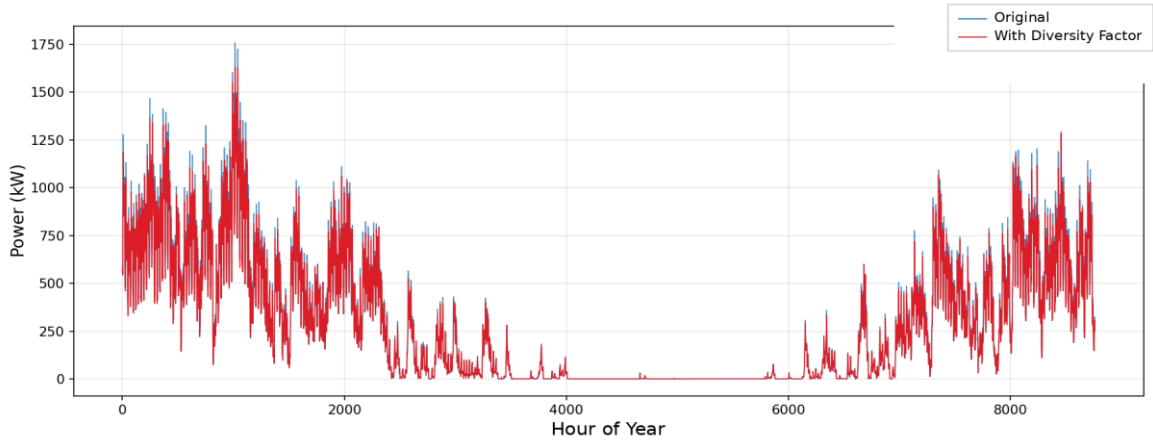


Figure 3: 23-building profile - annual time series, original vs. with DF ($a = 0.6$).

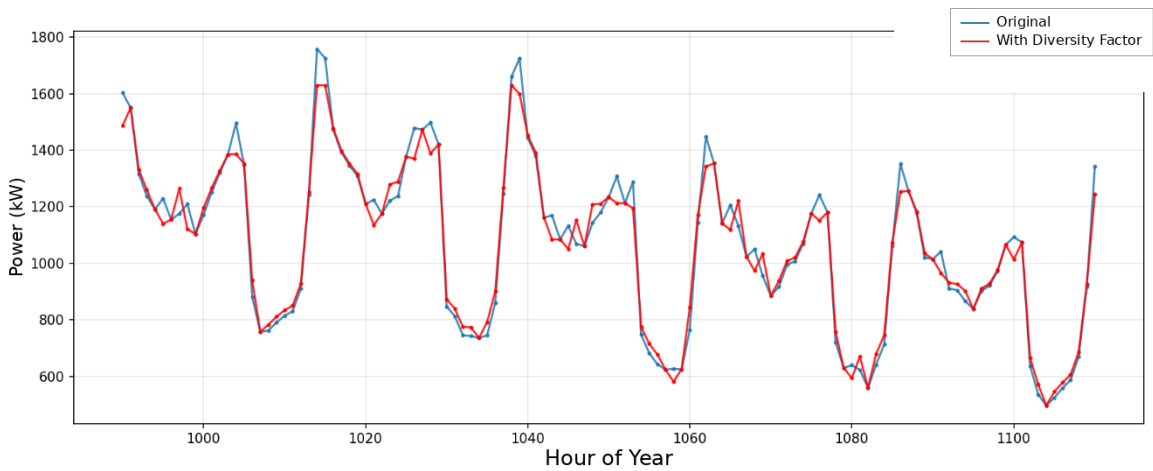


Figure 4: 23-building profile - detail view around the annual peak (hour 990-1110).

1.4 Assumptions and Limitations

- The diversity factor itself is not calculated from the time series but remains a freely selectable input parameter a ; c and d are fixed at Winter's original values, b is set to $1 - a$.
- Winter's formula is empirically validated only for $1 < n \leq 200$; for $n > 200$ (e.g. the 251-building case), the calculation is continued beyond the originally defined range.
- The underlying methodology is an approximation of the boundary conditions documented by nPro, not a confirmed reconstruction of the actual, non-publicly documented nPro algorithm.

1.5 Conclusion

The implemented Winter approach enables the application of a given DF to aggregated load profiles while satisfying the two boundary conditions: the peaks are reduced according to the given DF, while the total energy is preserved exactly. Energy redistribution takes place first locally, in the temporal vicinity of the respective load peaks, supplemented by a global correction if this is not sufficient to exactly satisfy both boundary conditions.

The approach can be applied both to already aggregated profiles and to multiple separate individual profiles. The resulting profiles show the expected peak reduction while the total sum remains unchanged.

The approach thus represents a functioning and traceable method for scaling load profiles with a given diversity factor. It should be understood as an independent implementation that satisfies the same boundary conditions derivable from the DF definition [2] but does not represent a reconstruction of the non-publicly documented internal nPro algorithm.

2 Shift Approach

This section describes a second, independent approach for modeling the simultaneity effect. The idea is to shift the load peaks of individual buildings in time, rather than clipping them directly and redistributing energy proportionally.

The goal of the Shift approach is to model the diversity effect through a physically traceable cause: real buildings do not heat in perfect synchrony, but with slight individual time offsets. If the individual profiles of multiple buildings are shifted individually and randomly in time before summation, the peak of the sum curve decreases due to the resulting decorrelation - without prescribing a target DF, unlike with the Winter formula.

2.1 Methodology

2.1.1 Determining the Time Shift per Building

Each of the n buildings receive an individual, random time offset, drawn uniformly within a time window. Rather than shifting all buildings uniformly or according to a fixed pattern, the time offsets are distributed randomly within the defined window, since this more realistically represents the real, non-uniform user behavior of different buildings.

2.1.2 Automatic Window Determination

The width of the shift window is a central, physically justified parameter. An excessively large shift of several hours would not be realistic - a heating load peak cannot be shifted arbitrarily far in time without losing its physical relationship to the actual load curve. So that the window does not have to be chosen manually for every number of buildings, an automatic saturation function dependent on n was defined:

$$window(n) = window_{max} \cdot (1 - e^{-\frac{n}{\tau}})$$

Here:

- $window(n)$ - the total width of the shift window used for n buildings, in hours,
- $window_{max}$ - the assumed maximum window width,
- τ - a parameter that determines how quickly the window approaches its maximum value,
- n - the number of buildings.

The saturation function models the assumed dependence of the shift window on the number of buildings. With few buildings, the window initially remains small, since only a small temporal spread of the individual load peaks is assumed. As the number of buildings increases, the window grows, since this accounts for a stronger temporal decoupling of the load peaks. For larger numbers of buildings, the increase increasingly flattens out, so that a further increase in n only slightly enlarges the shift window.

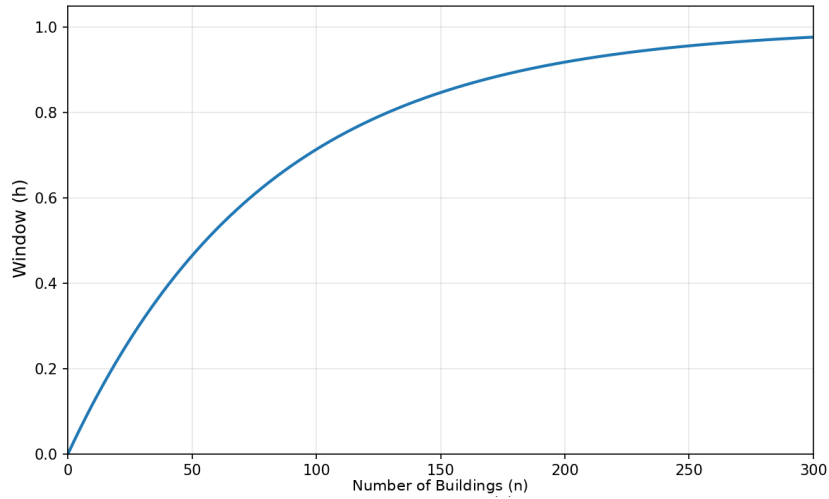


Figure 5: Dependence of the shift window width on the number of buildings

For the present approach, the following is assumed as a default upper bound of the shift window

$$window_{max} = 1h$$

For the parameter τ a value of

$$\tau = 80$$

was chosen as default as well. This choice was made so that the Shift approach functions well with $n \leq 200$.

With the parameters given above, the shift window develops as shown, for example, in Table 3.

Table 3

n	Share of maximum window	Shift window
10	0.118 h	≈ 7.1 min
20	0.221 h	≈ 13.3 min
50	0.465 h	≈ 27.9 min
100	0.713 h	≈ 42.8 min
150	0.847 h	≈ 50.8 min
200	0.918 h	≈ 55.1 min
250	0.956 h	≈ 57.4 min

For example, with 10 buildings only about 12% of the maximum possible window is used. With 50 buildings this is already about 47%, while with 200 buildings about 92% is reached. With 250 buildings, the window already reaches about 96% of the maximum value and thus amounts to about 57 minutes.

At $n=250$, the shift window thus already reaches about 96% of the maximum value. The Shift approach is therefore already nearly fully effective across the entire relevant range up to about 250 buildings, without the maximum shift already being reached at significantly smaller numbers of buildings.

For larger numbers of buildings, the window only grows marginally further and asymptotically approaches the maximum value of 1 hour. This keeps the influence of the number of buildings on

the shift window bounded and prevents unrealistically large temporal shifts from arising as the number of buildings increases.

2.1.3 Shifting the Individual Profile

The technical implementation of the shift depends on the temporal resolution of the input data (60, 15, 1 minute):

- Hourly input data (60 min): Since hourly values do not contain true sub-hourly information, the profile is first upsampled to a fine, periodic auxiliary grid (linear interpolation; the end of the year connects circularly to the start of the year). The shift is performed as an integer number of fine-grid steps, which preserves the sum exactly. The profile is then converted back to hourly resolution via block averaging, which preserves the annual energy exactly.
- High-resolution input data (15 or 1 min): No interpolation is needed here, since true sub-hourly information is already present. The shift is performed directly on the native time grid; the output resolution matches the input resolution - a user already working with 15-minute or minute profiles thus also obtains a result at this resolution, without rounding to hourly values.

2.2 Results

For two test cases with different individual building profiles generated individually with nPro [3] (10 and 50 buildings, respectively), the following picture emerges (Table 4):

Table 4

Metric	10 buildings	50 buildings
Time window (h / min)	0.118 h (~7.1 min)	0.465 h (~27.9 min)
Peak, raw (kW)	88.8	586.4
Peak, with Shift (kW)	88.1	580.5
Peak reduction (kW / %)	0.7 / 0.77 %	5.9 / 1.00 %
Annual total, raw (kWh)	76,561.2	505,001.9
Annual total, with Shift (kWh)	76,561.2	505,001.9

The annual energy is preserved exactly in both cases. The resulting peak reduction is less than 1.1% in both cases. From $n = 10$ to $n = 50$, the peak reduction increases only from 0.77% to 1.00%, while the shift window increases from 7.1 min to 27.9 min, nearly quadrupling.

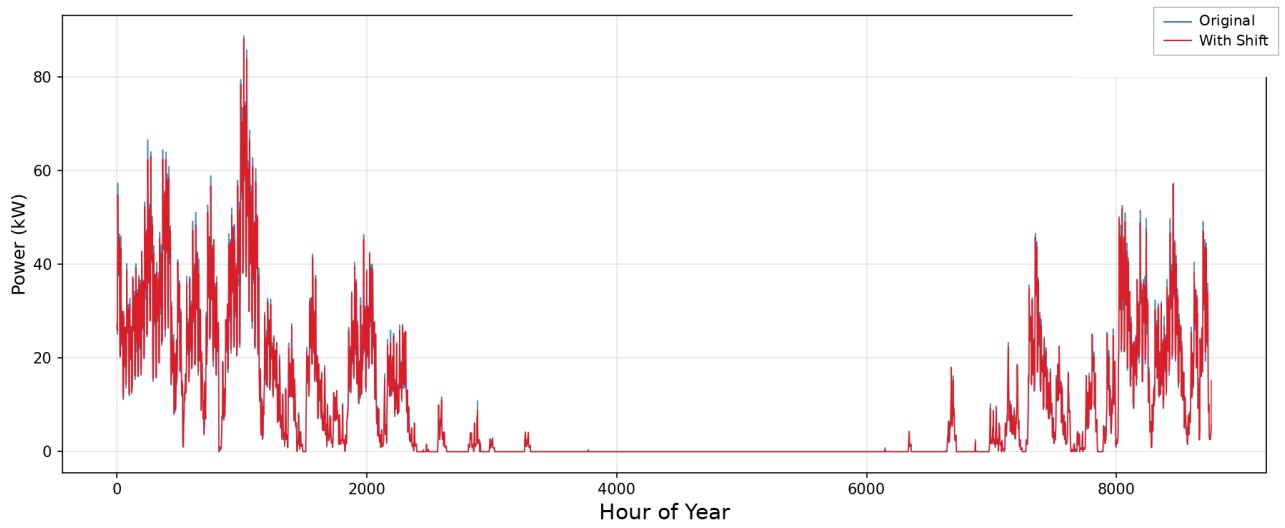


Figure 6: 10-building test case - annual time series, original vs. with Shift.

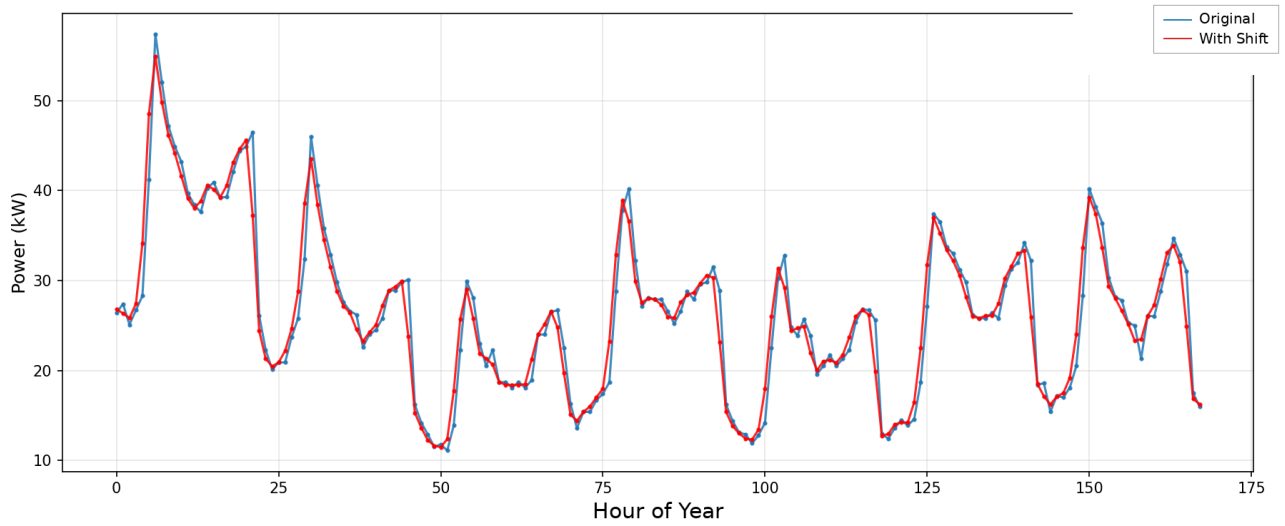


Figure 7: 10-building test case - detailed view of the first week of the year.

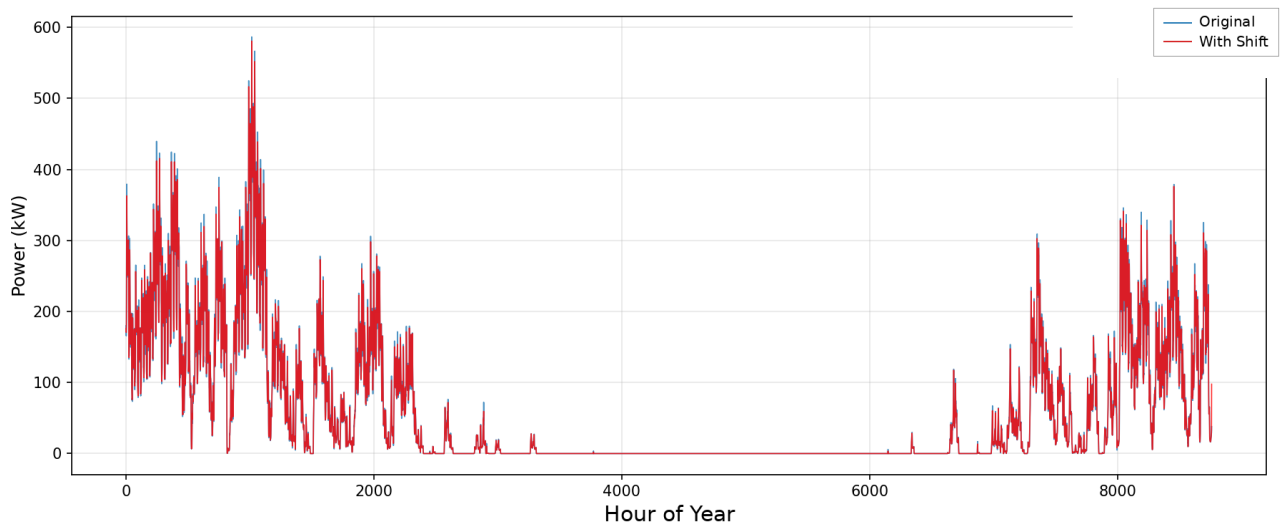


Figure 8: 50-building test case - annual time series, original vs. with Shift.

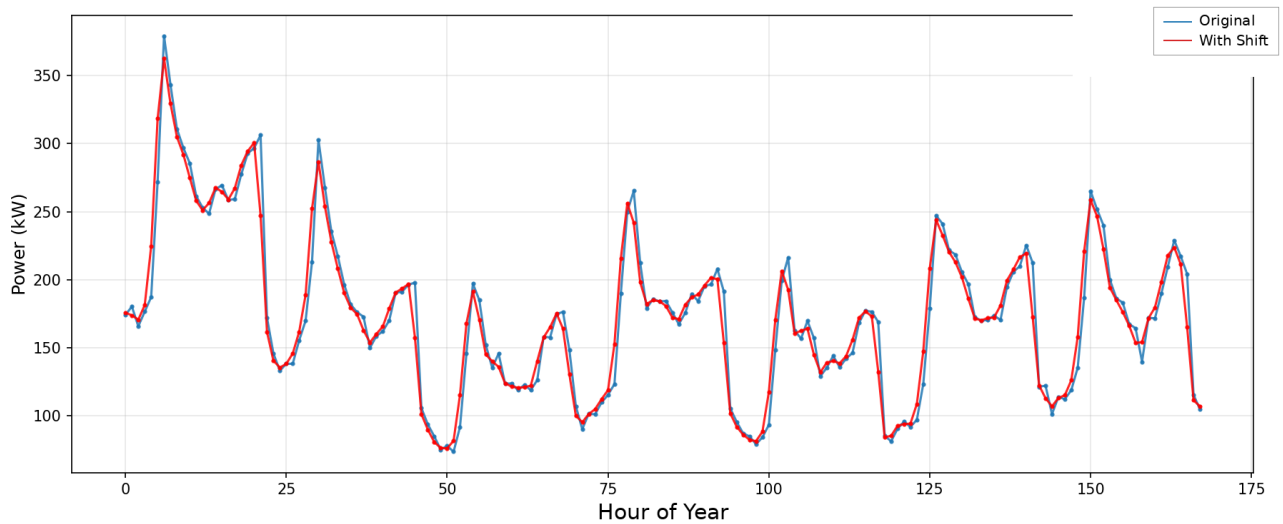


Figure 9: 50-building test case - detailed view of the first week of the year.

In the detailed view, the time-shift effect is clearly visible: at the peaks, the shifted curve visibly but only slightly deviates from the original curve, while the basic daily and weekly structure of the profile is preserved.

2.3 Assumptions and Limitations

- The approach can only be applied to separate individual building profiles, not to already aggregated profiles.
- $window_{max}$ and τ are both adjustable in the tool. The defaults ($window_{max} = 1h$ and $\tau = 80$) reflect the assumption that shifts up to one hour are realistic, and that the approach should cover an application range up to roughly 200 buildings - but users can change both values to fit their own use case.

2.4 Conclusion

The Shift approach represents an alternative mechanism for the simultaneity effect and preserves the annual energy per building exactly. Unlike the Winter approach, however, it leads, in the tested cases with 10 and 50 real buildings, only to a moderate peak reduction of under 1.1%. The reduction is therefore considerably lower than the DF-related reduction expected according to Winter et al.'s empirical formula for comparable numbers of buildings. The Shift approach should therefore be understood as a complementary method for investigating a single mechanism of the simultaneity effect, and not as an independently validated alternative for determining a simultaneity factor.

3 References

- [1] W. Winter, T. Haslauer, und I. Obernberger, „Untersuchungen der Gleichzeitigkeit in kleinen und mittleren Nahwärmenetzen," *Euroheat & Power*, Nr. 09–10, 2001.
- [2] nPro Energy GmbH, „Gleichzeitigkeitsfaktor in Wärmenetzen: Definition, Berechnung, Normen“. [Online]. Available: <https://www.npro.energy/main/de/knowledge/heat-networks/diversity-factor>. [Accessed: 03.09.2026].
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